

Reusing the same clipping shift

An extension of Hoza's Proposition 2.7, with $s = 1$

Following Hoza's notation [H], let $M : [w] \times \{0, 1\} \rightarrow [w]$, take $s = 1$ and $t = \log_2 n$, and assume $n \geq 2$ is a power of two and $1 \leq d \leq k$. All proposition numbers below refer to those notes. Assume state w is absorbing, as in Hoza's BPL reduction. If necessary, add a new absorbing last state and include it in w . Sample $h \sim H$ and $r \sim [2^d]$ uniformly and independently, once each. Use the notes' automaton clipping operation $\ominus_d$, including its canonical reconstruction, and pad alphabets to k bits as in the notes. Define

$$\begin{aligned}\widehat{M}^{(0)} &= M, & \widehat{M}^{(i)} &= (\widehat{M}^{(i-1)})_h \ominus_d r, \\ M^{(0)}(r) &= M, & M^{(i)}(r) &= (M^{(i-1)}(r))^2 \ominus_d r.\end{aligned}\tag{1}$$

The second sequence is only for analysis. For fixed r , it is independent of h . As in the notes, norms and matrix entries refer to transition probability matrices.

Claim. Except with probability

$$O(nw^3 2^{-d}) + w^5 \log n \cdot 2^{O(d)} \cdot 2^{-k}$$

over (h, r) , the output satisfies $\|\widehat{M}^{(t)} - M^n\|_1 \leq 4nw 2^{-d}$.

Proof. The only new part of Proposition 2.7 is the estimate for bad clipping shifts. Its deterministic error bound and its argument over h can be reused.

1. *Reuse the deterministic error bound.* The induction at the end of the proof of Proposition 2.7 does not require independent shifts. With $s = 1$ it gives, for every r and every i ,

$$\|M^{(i)}(r) - M^{2^i}\|_1 \leq 4w(2^i - 1) 2^{-d}.\tag{2}$$

2. *Count the bad values of the shared shift.* Fix a round $i \in [t]$ and states $u, v < w$, and write

$$p(r) = \left((M^{(i-1)}(r))^2 \right)_{u,v}, \quad p^* = (M^{2^i})_{u,v}.$$

On the block indexed by $[w - 1]$, clipping only decreases entries, and the absorbing last state never sends probability back into this block. The scalar operation $p \ominus_d r$ is nondecreasing in p and nonincreasing in r , while squaring nonnegative matrices preserves entrywise order. Induction therefore shows that $p(r)$ is nonincreasing in r and $p(r) \leq p^*$.

Combining (2) with the powering inequality from the same proof gives

$$p^* - 4w(2^i - 2) 2^{-d} \leq p(r) \leq p^*.\tag{3}$$

Call r bad for (i, u, v) if $p(r) - r 2^{-2d}$ is at distance less than 2^{-2d-1} from an integer multiple of 2^{-d} . This is exactly the boundary condition used in the proof of Proposition 2.2, now applied to the varying input $p(r)$.

Set $g(r) = p(r) - r2^{-2d}$. Monotonicity gives

$$g(r) - g(r+1) \geq 2^{-2d}. \quad (4)$$

Thus each fixed rounding boundary is within distance *strictly less than* 2^{-2d-1} of $g(r)$ for at most one r . By (3), the range of g lies in an interval of length

$$(4w(2^i - 2) + 1) 2^{-d}.$$

Extending each end by 2^{-2d-1} yields an interval containing at most $4w(2^i - 2) + 3$ multiples of 2^{-d} . Hence at most that many shifts are bad for (i, u, v) .

Let R_{bad} be the union of these bad sets over $i \in [t]$ and $u, v < w$. Since $\sum_{i=1}^t (2^i - 2) = 2(n - 1 - t)$,

$$\Pr_r[r \in R_{\text{bad}}] \leq w^2 2^{-d} \sum_{i=1}^t (4w(2^i - 2) + 3) = O(nw^3 2^{-d}). \quad (5)$$

This is a union bound on subsets of the single sample space $[2^d]$; no independence between rounds is used.

For $r \notin R_{\text{bad}}$, the proof of Proposition 2.2 makes every non-sink entry at every ideal step stable under perturbations of magnitude less than 2^{-2d-1} . Agreement of these clipped entries determines the last column by row sums. The absorbing row is fixed. Thus the conclusion needed from Proposition 2.4 holds along the entire ideal sequence.

3. *Finish as in Proposition 2.7.* Fix $r \notin R_{\text{bad}}$. The ideal automata $M^{(0)}(r), \dots, M^{(t-1)}(r)$ are now fixed independently of h . The fixed-sequence hash estimate in the proof of Proposition 2.7, specialized to $s = 1$, gives

$$\left\| (M^{(i-1)}(r))_h - (M^{(i-1)}(r))^2 \right\|_1 < 2^{-2d-1} \quad \text{for all } i \in [t], \quad (6)$$

except with probability $w^5 \log n \cdot 2^{O(d)} \cdot 2^{-k}$ over h .

Assuming (6), induction in (1) gives $\widehat{M}^{(i)} = M^{(i)}(r)$ as automata: equality at round $i - 1$, followed by (6) and clipping stability, gives equality at round i . Canonical reconstruction is included in Θ_d , so equality holds for transition functions, not only for matrices.

Finally, (2) at $i = t$ gives the claimed error. Combining (5) with the conditional failure probability over h proves the claim. There is no union bound over all values of r in the hash estimate. $\square$

For $\varepsilon \in (0, 1)$, suitable $d, k = O(\log(wn/\varepsilon))$ give error at most ε with probability at least $2/3$.

Reference. [H] William Hoza, *BPL $\subseteq$ SC and the Saks–Zhou theorem (lecture notes)*, Winter 2025. Section 2, especially Propositions 2.2–2.4 and the proof of Proposition 2.7.